

Rajiv Gandhi University of Health Sciences, Karnataka

Fourth Semester B. Pharm Degree Examination – 29-Oct-2024

Time: Three Hours

Max. Marks: 75 Marks

Physical Pharmaceutics - II

Q.P. CODE: 5015

Your answers should be specific to the questions asked

Draw neat labeled diagrams wherever necessary

All the Questions are compulsory.

LONG ESSAYS

2 x 10 = 20 Marks

1. What are colloids? Explain in detail properties of colloids.

OR

Define Thixotropy. Explain the different methods to determine the thixotropic behaviour of liquids. Write a note on its applications in pharmacy.

2. Define suspensions. Explain in detail all the parameters that have to be considered in the formulation of suspensions.

SHORT ESSAYS

7 x 5 = 35 Marks

3. With the help of neat labelled diagram explain the principle involved in the working of Ostwald's viscometer.

OR

Explain in detail air permeability method for the determination of specific surface area.

4. Define Arrhenius plot and give its significance in calculation of shelf life.

OR

Discuss protection of colloids with suitable example.

5. Explain plastic and pseudoplastic flow.
6. Note on particle size determination by microscopic and sedimentation method.
7. Discuss in detail accelerated stability testing.
8. What are hydrophobic colloids? Explain any two methods of its preparation.
9. Enumerate different methods for determination of true density. Explain any one method in detail.

SHORT ANSWERS

10 x 2 = 20 Marks

10. Define shelf life of a medicinal product.
11. What do you mean by degree of flocculation?
12. Define porosity and write any two applications.
13. State Hardy schulze rule
14. Write any two applications of microemulsions.
15. Write any two methods for the prevention of photolytic degradation.
16. Define yield value.
17. Write the formula for average particle size.
18. Classify disperse systems.
19. Give Stoke's law.
